

Customer Shopping Behavior Analysis

Data Analysis Report — Kesar Deaulkar

PYTHON

SQL SERVER

POWER BI





Executive Summary

Purpose

Analyze purchasing trends, revenue, discounts, subscriptions & product performance

Approach

Python cleaning » SQL querying » Power BI dashboards

Impact

Improve marketing, optimize inventory, personalize experiences, grow revenue

Project Objectives

01

Analyze purchasing behavior

02

Revenue by customer segment

03

Discount impact evaluation

04

Subscriber vs non-subscriber
spend

05

Top-rated & frequent products

06

Customer loyalty patterns

07

Interactive dashboard

DATASET

3,900 Records · 18 Attributes

Customer ID	Customer Name	Email	Phone	City	State	Join Date	Customer Segment	Total Orders	Total Spend (USD)	Average Order Value (USD)	Last Order Date	Customer Status	Satisfaction Score (1-5)	
 1	C001	Emma Johnson	emma.johnson@email.com	(555) 123-4567	New York	NY	2023-01-15	Premium	28	\$4,562.30	\$162.94	2024-05-20	Active	★★★★★ 4.8
 2	C002	Liam Smith	liam.smith@email.com	(555) 234-5678	Los Angeles	CA	2023-02-20	Standard	15	\$2,145.60	\$143.04	2024-05-18	Active	★★★★☆ 4.2
 3	C003	Olivia Williams	olivia.williams@email.com	(555) 345-6789	Chicago	IL	2023-03-10	Premium	32	\$5,672.80	\$177.28	2024-05-21	Active	★★★★★ 4.9
 4	C004	Noah Brown	noah.brown@email.com	(555) 456-7890	Houston	TX	2023-03-25	Standard	12	\$1,654.35	\$137.86	2024-05-17	Active	★★★★☆ 4.1
 5	C005	Ava Davis	ava.davis@email.com	(555) 567-8901	Phoenix	AZ	2023-04-05	Basic	8	\$876.40	\$109.55	2024-05-10	Active	★★★☆☆ 3.3
 6	C006	William Miller	william.miller@email.com	(555) 678-9012	Philadelphia	PA	2023-04-18	Standard	18	\$2,763.50	\$153.53	2024-05-19	Active	★★★★☆ 4.3
 7	C007	Sophia Wilson	sophia.wilson@email.com	(555) 789-0123	San Antonio	TX	2023-05-01	Premium	25	\$4,121.75	\$164.87	2024-05-20	Active	★★★★☆ 4.6
 8	C008	James Moore	james.moore@email.com	(555) 890-1234	San Diego	CA	2023-05-12	Standard	14	\$1,890.20	\$135.01	2024-05-16	Inactive	★★★☆☆ 3.0
 9	C009	Isabella Taylor	isabella.taylor@email.com	(555) 901-2345	Dallas	TX	2023-06-03	Premium	30	\$4,987.90	\$166.26	2024-05-21	Active	★★★★★ 4.7
 10	C010	Benjamin Anderson	benjamin.anderson@email.com	(555) 012-3456	San Jose	CA	2023-06-15	Standard	17	\$2,345.80	\$137.99	2024-05-18	Active	★★★★☆ 4.4
 11	C011	Mia Thomas	mia.thomas@email.com	(555) 123-5670	Austin	TX	2023-07-01	Basic	6	\$654.25	\$109.04	2024-05-09	Inactive	★★★☆☆ 3.1
 12	C012	Elijah Jackson	elijah.jackson@email.com	(555) 234-6780	Jacksonville	FL	2023-07-11	Standard	13	\$1,753.60	\$134.89	2024-05-15	Active	★★★★☆ 4.0
 13	C013	Charlotte White	charlotte.white@email.com	(555) 345-7890	Fort Worth	TX	2023-07-20	Premium	27	\$4,432.10	\$164.15	2024-05-20	Active	★★★★☆ 4.5
 14	C014	Alexander Harris	alexander.harris@email.com	(555) 456-8901	Columbus	OH	2023-08-02	Standard	11	\$1,432.50	\$130.23	2024-05-14	Inactive	★★★☆☆ 2.9
 15	C015	Amelia Martin	amelia.martin@email.com	(555) 567-9013	Charlotte	NC	2023-08-14	Premium	29	\$4,812.60	\$165.95	2024-05-21	Active	★★★★★ 4.8
 16	C016	Daniel Thompson	daniel.thompson@email.com	(555) 678-0124	Indianapolis	IN	2023-08-25	Basic	7	\$743.30	\$106.19	2024-05-08	Active	★★★★☆ 3.4
 17	C017	Harper Garcia	harper.garcia@email.com	(555) 789-1235	Seattle	WA	2023-09-05	Standard	16	\$2,234.40	\$139.65	2024-05-17	Active	★★★★☆ 4.2
 18	C018	Lucas Martinez	lucas.martinez@email.com	(555) 890-2346	Denver	CO	2023-09-12	Premium	26	\$4,321.70	\$166.22	2024-05-19	Active	★★★★☆ 4.6
 19	C019	Evelyn Robinson	evelyn.robinson@email.com	(555) 901-3457	Nashville	TN	2023-09-20	Standard	10	\$1,298.75	\$129.88	2024-05-13	Inactive	★★★☆☆ 3.0
20	C020	Sebastian Clark	sebastian.clark@email.com	(555) 012-4567	Miami	FL	2023-10-01	Basic	5	\$542.15	\$108.43	2024-05-07	Active	★★★☆☆ 3.2

Demographics

Age, Gender, Location

Purchase

Amount, Category, Item

Behavior

Frequency, Reviews, Subscription

Data Preprocessing (Python)

Data Cleaning

- Loaded via Pandas, checked structure & missing values
- Median-filled Review Ratings by category
- Lowercase columns, replaced spaces with underscores
- Removed redundant **Promo Code Used** column

Feature Engineering

Age Group

Young Adult · Adult ·
Middle-aged · Senior

Frequency (Days)

Weekly=7 ·
Fortnightly=14 ·
Monthly=30

SQL Business Analysis



Revenue by Gender



High-Spending Discount Customers



Top 5 Rated Products



Shipping Analysis



Subscriber vs Non-Subscriber



Customer Segmentation



Top Products by Category



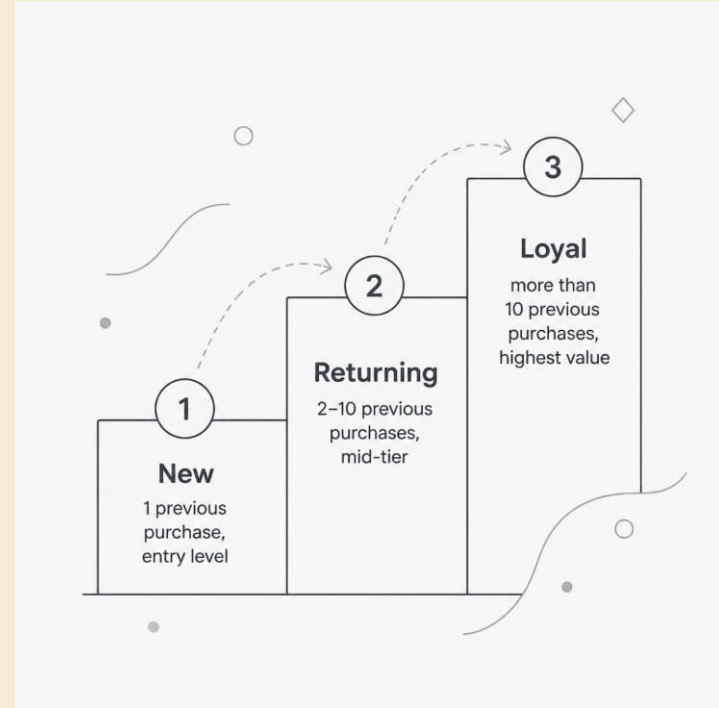
Repeat Buyer Subscription



Revenue by Age Group

Customer Segmentation

Customers classified by previous purchase count to support targeted marketing.



Power BI Dashboard

Revenue

Overview, by Gender, by Age Group

Customers

Distribution, Segmentation, Subscription

Products

Category Performance, Ratings

Transactions

Shipping & Discount Analysis



Key Insights & Recommendations

Key Insights

Spending differs across demographics

Subscribers drive significant revenue

Loyal customers account for major purchases

Discounts influence select products only

Recommendations

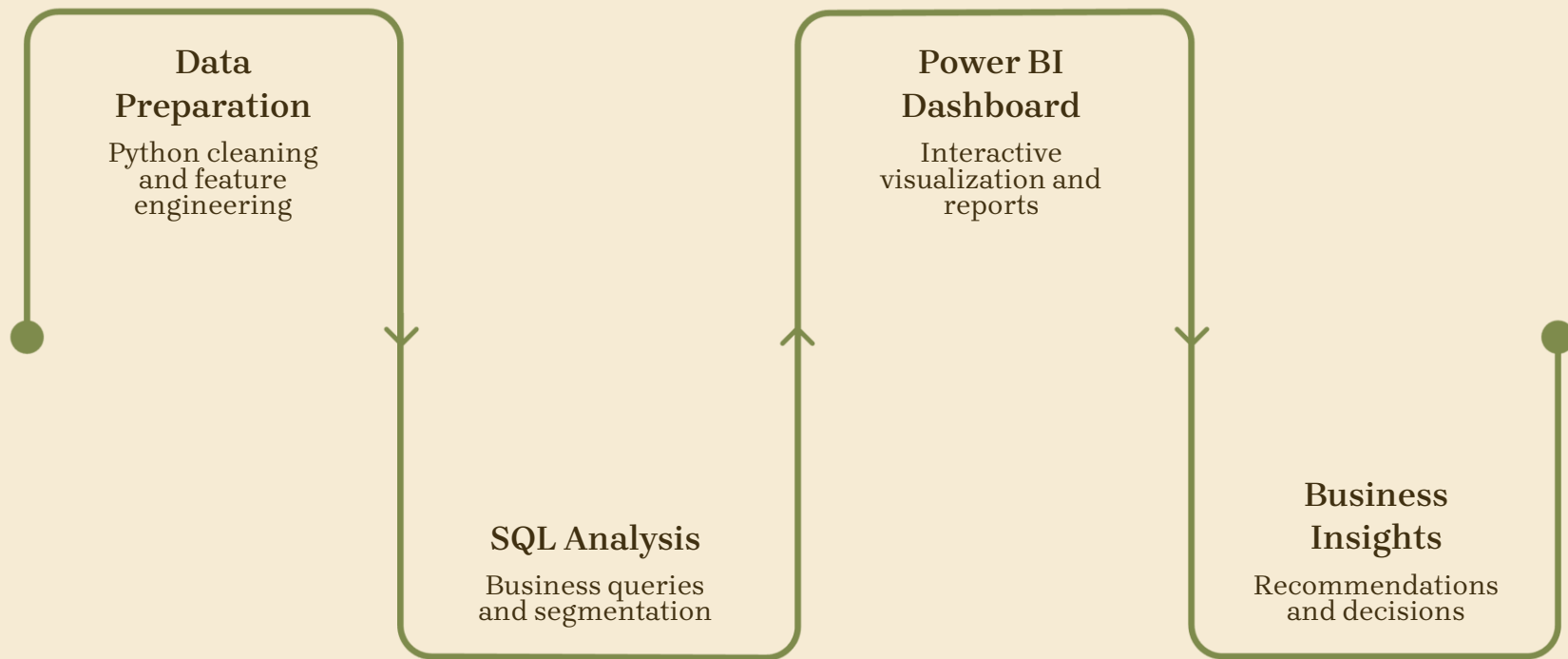
Target loyal & returning customers

Expand subscription programs

Personalize by age & history

Reduce discounts on high-demand products

Project Workflow & Tech Stack



End-to-end analytics pipeline from raw data to actionable business recommendations.

Python · Pandas · NumPy
Cleaning & engineering

SQL Server (SSMS)
Business queries

Matplotlib · Power BI
Visualization & dashboards